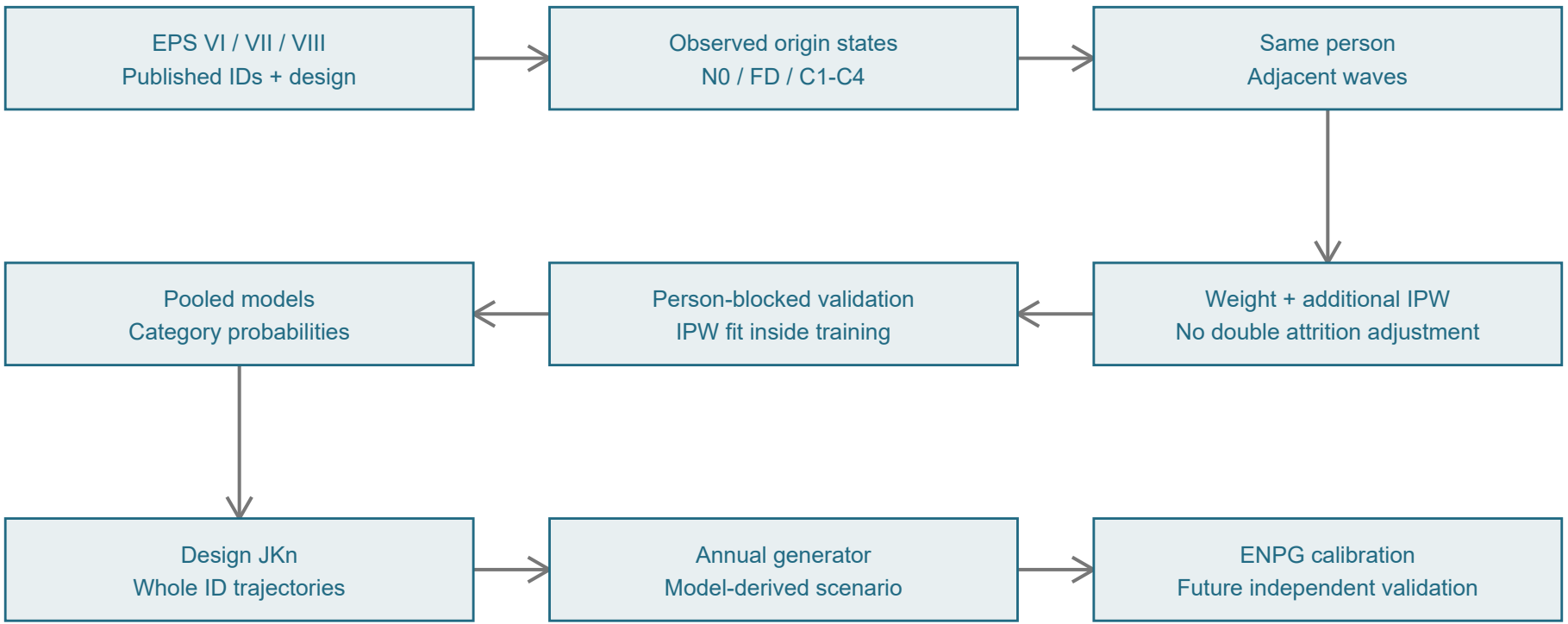From survey responses to transition probabilities